RealQM and the Electron Gas: what extends, what does not, and what a zero-temperature pressure would require

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Abstract

RealQM models matter as non-overlapping unit charge densities in ordinary three-dimensional space, one per electron, meeting at free boundaries and minimising the Coulomb energy. It has been developed and tested on *bound* matter — atoms, molecules and nuclei — where an attractor is always present. This note asks what happens when the framework is carried to a domain it was not built for: the electron gas, and with it plasmas, metals and dense matter. The answer separates cleanly. The **form** of the theory extends exactly: partitioning a gas of density n into non-overlapping cells gives a kinetic energy density $Cn^{5/3}$ — the Thomas–Fermi form, obtained here from non-overlap rather than from Fermi statistics — and everything that follows from charge and inertia is reproduced without adjustment, the plasma frequency $\omega_p^2 = 4\pi n$ exactly. What does *not* extend is the coefficient: for a uniform gas with the free-boundary interface used throughout RealQM, the minimising density is constant, so $C = 0$. The model then has no pressure at zero temperature, hence no screening length and no plasmon dispersion, and predicts that a cold dense electron gas exerts no pressure at all. Since a metal at room temperature is already degenerate ($T/T_F \approx 10^{-2}$), this is a definite conflict with observation, and one that thermal motion cannot repair: temperature supplies a pressure $\sim nk_BT$ which vanishes as $T \to 0$, while the missing one does not. The obvious repair — a Robin interface condition $\psi' + \beta\psi = 0$ — reproduces Thomas–Fermi exactly at the single dimensionless value $\beta a = 1.146$, but costs helium 0.19 Ha, some 20% of its ionisation energy, so it cannot be adopted consistently. We conclude that extending RealQM beyond bound matter requires a genuinely new ingredient: a mechanism by which compressing a partition costs energy at zero temperature. That is stated here as a well-posed open problem rather than a defect, since it lies outside the domain in which the framework has so far been tested.

1 Why the electron gas is a new domain

RealQM [1, 2] represents an N-electron system by N non-negative densities ψ_i on non-overlapping domains Ω_i, with energy

$$E[\{\psi_i\},\{\Omega_i\}] = \sum_i \int_{\Omega_i} \Big(\tfrac{1}{2}|\nabla\psi_i|^2 - \sum_a \frac{Z_a}{|x-x_a|}\psi_i^2\Big)dx + \tfrac{1}{2}\sum_{i\neq j}\iint \frac{\psi_i^2(x)\psi_j^2(y)}{|x-y|}\,dx\,dy, \qquad (1)$$

subject to $\int_{\Omega_i}\psi_i^2 = 1$, the interfaces free to move to where the energy says. There is no wavefunction in $3N$ dimensions, no antisymmetry and no self-interaction; the role played by the exclusion principle is played instead by the requirement that the domains do not overlap.

Every test of this construction so far — ionisation energies across the periodic table, molecular binding, nuclear binding by dual Coulomb confinement — shares one feature: an *attractor* is present. A nucleus is always there to shape the density. The question this note asks is what happens when it is not, which is the situation in an electron gas, a plasma, or a metal, and which is a domain RealQM was neither built for nor tested on. It should not be surprising if new ingredients are required, and the purpose here is to identify exactly which.

2 The form extends: non-overlap gives the Thomas–Fermi scaling

Partition a gas of density n into non-overlapping cells, one per electron, of linear size $a = n^{-1/3}$. The localisation cost of a cell scales as $\hbar^2/2m\,a^{-2} \propto n^{2/3}$, so the kinetic energy density is

$$t(n) = C\,n^{5/3}, \qquad (2)$$

with C fixed by the shape of the cell and the condition at its faces. This is precisely the Thomas–Fermi form, and it is worth pausing on where it came from: not from Fermi statistics, but from the geometry of a partition. The framework's central premise — that exclusion is a statement about regions rather than about exchange symmetry — delivers the correct *scaling* of the degenerate electron gas immediately.

Everything downstream then follows from C. Minimising (1) at fixed chemical potential with a test charge Q present, and linearising about the uniform state, gives $\delta n = (9n_0^{1/3}/10C)\varphi$ and hence, with Poisson's equation, a Helmholtz problem $\nabla^2\varphi - k^2\varphi = -4\pi Q\delta(x)$ with

$$k^2 = \frac{18\pi\,n_0^{1/3}}{5C}, \qquad (3)$$

so that $\varphi = Q\,e^{-kr}/r$: Yukawa screening, with $\lambda = 1/k \propto n^{-1/6}$. Setting C to the Thomas–Fermi coefficient $C_{\mathrm{TF}} = \frac{3}{10}(3\pi^2)^{2/3} = 2.8712$ reproduces $k_{\mathrm{TF}}^2 = (4/\pi)(3\pi^2 n)^{1/3}$ *exactly*, at every density (Table 1).

n (a.u.)	λ from (3) with C_{TF}	λ_{TF}
0.01	$1.086\,a_0$	$1.086\,a_0$
0.1	$0.740\,a_0$	$0.740\,a_0$
1.0	$0.504\,a_0$	$0.504\,a_0$

Table 1: The form is exact. Everything rests on the single constant C.

3 What extends without adjustment: the plasma frequency

Displace the electron distribution rigidly by ξ against the ion background. The resulting surface charge produces a restoring field $4\pi n\xi$, so $\ddot{\xi} = -4\pi n\xi$ and

$$\omega_p^2 = 4\pi n, \tag{4}$$

free of temperature and of any statistical assumption. **RealQM reproduces this exactly**, and for a structural reason: under a rigid translation each ψ_i is carried along unchanged, so $\nabla\psi_i$ is unchanged, the localisation term contributes nothing, and what remains is Coulomb and inertia — which is all the result depends on. No property of C enters.

The same argument identifies what RealQM should do about **Landau damping**. The framework is not a single fluid: N electrons are N domains, each carrying its own current and hence its own drift velocity, which is structurally a multi-beam plasma. Multi-beam models reproduce Landau damping as phase mixing among cold streams, and do so reversibly, in agreement with plasma-echo experiments. Where the velocity spread is thermal, the framework should therefore give collisionless damping for the right reason rather than by an inserted term.

4 What does not extend: the coefficient is zero

The step from (2) to a number requires the cell to have a nonzero localisation cost, and with RealQM's own interface condition it does not.

The kinetic term in (1) is integrated over each electron's *own* domain, and no node is imposed where domains meet; the calibration of atomic RealQM found the free (Neumann) plane decisive against a Dirichlet node. Take then $\psi_i \equiv |\Omega_i|^{-1/2}$, constant on each cell. This satisfies the normalisation, makes the total density $\sum \psi_i^2$ uniform, minimises the electrostatic energy against a neutralising background, and has $\nabla\psi_i = 0$ identically. The kinetic energy is exactly zero, and the discontinuity at the cell face costs nothing because the integral runs over Ω_i alone.

The uniform state is therefore the global minimum, with

$$C = 0 \tag{5}$$

at every density. By (3) the screening length is infinite. The compressibility correction to the plasma frequency, $\omega^2 = \omega_p^2 + \frac{3}{5}k^2 v_F^2$, vanishes, so the predicted plasmon is dispersionless. And the pressure of a cold electron gas is zero.

Two attempts to locate a missing term fail. Allowing the interfaces to move changes nothing:

with constant ψ_i the kinetic energy remains zero and the electrostatics merely fixes the cells to equal volume. Requiring each domain to be connected and to carry unit charge changes nothing either, since any equal-volume connected tiling admits the same constant solution. The result is a property of the functional, not an oversight in its evaluation.

5 The interface condition cannot repair it

Since the Neumann condition gives $C = 0$ and a hard wall gives $C = 3\pi^2/2 = 14.804$, the two bracket the target, which suggests a Robin condition $\psi' + \beta\psi = 0$ at the interface. For a cubic cell of side a the problem separates, with $\kappa = ka$ solving $\kappa \tan(\kappa/2) = \beta a$ and $C = \frac{3}{2}\kappa^2$:

interface condition	κ	C	C/C_{TF}
Neumann, $\beta a = 0$ (present choice)	0	0	0.00
Robin, $\beta a = 1.146$	1.3835	2.8712	**1.00**
Dirichlet, $\beta a \to \infty$	π	14.804	5.16

Table 2: One dimensionless number reproduces Thomas–Fermi exactly — and with it the screening length, the plasmon dispersion and the zero-temperature pressure, since all three follow from C.

This is not adopted here, for a reason established by direct test. Helium in RealQM is two electron domains meeting at a midplane through the nucleus. Solving that problem on a grid with the interface condition varied, at otherwise identical numerics, gives

β (a_0^{-1})	E (Ha)	T per electron	shift
0 (Neumann)	−2.0224	1.887	—
0.25	−2.0067	1.726	+0.016
1.146	−1.8358	1.164	**+0.187**
20 (near Dirichlet)	−0.9608	0.350	+1.062

Table 3: Helium is strongly sensitive to the interface condition. The absolute energies are from a coarse grid and are not converged; the comparison is between rows at identical numerics.

The value the gas requires costs helium 0.19 Ha, about 20% of its ionisation energy. The reason is geometric and does not depend on the accuracy of the calculation: in helium the interface plane passes *through the nucleus*, where the density is greatest, so any departure from Neumann suppresses density exactly where binding is won. The electron responds by spreading away from the plane, which is why the kinetic energy falls as the total energy rises.

The situation is therefore symmetric and unfavourable. In a uniform gas the density is constant, so the interface condition is the only place physics could enter — and Neumann makes it contribute nothing. In an atom the interface passes through the density maximum, so the condition is decisive — and only Neumann leaves the atom alone. *The interface is inconsequential exactly where the gas needs it to matter, and decisive exactly where the atom needs it not to.* No single β serves both, and a value that switched between regimes could not be applied at all, since partially ionised plasmas and metals contain bound and free electrons in the same system with no surface between them.

6 The role of temperature, and why it does not close the gap

At finite temperature the domains move and fluctuate in shape, the density within a cell is no longer constant, and $\nabla\psi_i \neq 0$: there is a kinetic energy, and with it a pressure. RealQM should reproduce it, since it follows from charge, inertia and thermal motion, all of which the framework carries. But this pressure scales as nk_BT and vanishes as $T \to 0$.

The missing pressure does not. Splitting the two contributions:

contribution	survives $T \to 0$?	RealQM
thermal, $\sim nk_BT$	no	should be reproduced
degenerate, $\sim nE_F$	**yes**	**zero**

and the second is not an exotic regime. A metal at room temperature has $T_F \sim 10^4$–10^5 K, so $T/T_F \approx 10^{-2}$: sodium and aluminium are held up almost entirely by the pressure that survives at zero temperature. The conflict is therefore blunt and requires no appeal to any formalism: *the model predicts that a cold dense electron gas exerts no pressure, and sodium exists.* The same statement applies to white dwarf support and to the interiors of dense stars.

It is worth being clear about what is and is not being claimed. Antisymmetry of a many-body wavefunction is a formalism and is not itself observed; the framework is entitled to account for the phenomena by other means, and that is its programme. What is observed — the mass–radius relation of white dwarfs, the linear electronic specific heat, de Haas–van Alphen oscillations, the bulk moduli of simple metals, positive plasmon dispersion — is that a large pressure exists at temperatures far below T_F. A theory of matter must produce it somehow.

7 What an extension would require

The open problem can be stated in the framework's own terms, without reference to exchange symmetry:

> *What geometric property of a partition should cost energy when its cells shrink?*

Non-overlap alone does not, because a constant density in a cell has no gradient however small the cell. Yet compressing a metal demonstrably costs energy. Something in the description of a partition must resist compression at zero temperature, and identifying it is what carrying RealQM beyond bound matter requires.

Two remarks may help locate it. First, the failure is specific to the *uniform* case: wherever an attractor shapes the density, the profile itself supplies the gradients, which is why the atomic and molecular results are unaffected and why the deficiency was not visible before. Second, the correct coefficient lies between the two principled interface conditions and is produced by neither (0 and 5.16 against a required 1.00), which is the signature of a missing mechanism rather than of a mis-set parameter.

Until such a mechanism is found, the honest statement of scope is that RealQM is a theory of **bound** matter — atoms, molecules and nuclei, where an attractor is present — and that its extension to free-electron matter awaits an ingredient it does not yet contain. Nothing in the established results depends on the extension, and stating the boundary costs the framework nothing it had.

A note on what was tested numerically

The screening relation (3), the vanishing of C, the plasma frequency and the Robin table are analytic. Table 3 is a direct grid solution of helium as two half-space domains with the interface condition varied, and its absolute energies (-2.02 Ha against the experimental -2.9037) reflect a coarse mesh; only the differences between rows are used. A separate calculation of ionisation potential depression in dense plasma reproduces Stewart–Pyatt in the dilute limit (3.61 against 3.57 eV at $n = 10^{21}$ cm^{-3}) but becomes unphysical at high density, where a smeared neutralising background can no longer stand in for explicit neighbouring domains; no conclusion is drawn from it here. Scripts are available with the source.

References

[1] C. Johnson, *RealQM: Quantum Mechanics as Multiphase 3D Continuum Mechanics*, in [2].

[2] C. Johnson, *Real Quantum Mechanics* (Body&Soul, Vol. VI), `https://claes542.github.io/RealMolecule/realquantum3.pdf`.

[3] C. Johnson, *RealNucleus: Nuclear Binding as Dual Coulomb Confinement, without a Strong or Weak Force*, in [2].